When Correlation Lies: Stress-Testing Pairwise Net Influence for Detecting Coordination Among Agents[1]

Track 1.4: Markets and Collusion | Apart Research AI Collusion Research Sprint (October 23–25, 2026)

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Verified Pipeline: Python 3.13 / CPU Benchmarks & Modal CPU Cloud ($5.41 Promotional Credit, $0.00 Out-of-Pocket Spend)

Evaluations: Canonical Scripted: 260 trajectories (54,000 rounds) | Open-Weight LLM: 99 trajectories (9,180 decisions)

ABSTRACT

As autonomous AI agents are increasingly deployed in high-frequency trading, automated procurement, and algorithmic pricing, covert collusion poses severe economic and systemic risks. Information-theoretic detectors, notably Pairwise Net Influence (PNI; Bonjour et al., 2022), measure directed action dependence via Conditional Mutual Information (CMI) conditioned on observable public game state. However, whether PNI can distinguish deliberate multi-agent coordination from spurious correlation induced by shared context remains an open empirical question. In this work, we stress-test PNI against unconditioned marginal mutual information and raw conditional mutual information baselines in an iterated 4-agent game across four controlled conditions: independent play, shared-context confounding, shared-private-signal coordination, and coordination-strength sweeps. Across 20 independent seeds and trajectory lengths (T in [50, 500]), unconditioned mutual information exhibits a 38-fold surge under shared context, falsely accusing independent agents of collusion (FPR = 1.000). Conditioning on observable state eliminates this contextual surge (CMI FPR = 0.000), while PNI competitor subtraction isolates coordinating pairs with 100% Top-1 accuracy (Wilson 95% CI: [0.839, 1.000], MRR = 1.000). Furthermore, expanding evaluations to 9,180 open-weight LLM decisions (Qwen2.5-1.5B) across horizons T in {5, 20, 50} reveals an empirical finite-sample sparsity effect: ultra-short horizons (T=5) attenuate CMI ranking accuracy (66.7%), whereas sufficient history (T >= 20) restores 100% Top-1 identification accuracy (25/25 seeds). Our findings prove state conditioning is indispensable for behavioral auditing while highlighting sample horizon boundaries.

1. Introduction

Autonomous algorithmic agents increasingly govern financial exchanges, automated multi-unit auctions, and dynamic e-commerce pricing. A critical frontier in AI safety is preventing **algorithmic collusion**—scenarios where autonomous agents synchronize strategies to extract supracompetitive rents at the expense of human market participants (Calvano et al., 2020; Schwalbe, 2018). Unlike traditional cartels that rely on explicit communication logs, AI agents can establish implicit coordination purely through tacit observational learning or private coordination signals. Consequently, behavioral auditing tools that operate solely on external observational logs are essential for oversight.

In Track 1.4 (Markets and Collusion) of the AI Collusion Research Sprint, we address the core research question: *Can information-theoretic approaches detect collusion in multi-agent games?* Specifically, we stress-test **Pairwise Net Influence (PNI)** (Bonjour, Aggarwal, & Bhargava, UAI 2022). While PNI is theoretically sound, real-world market auditing confronts a severe confounder: **shared-context confounding**. When non-coordinating agents independently react to identical market histories, price shocks, or public order books, their actions naturally synchronize. If auditing metrics mistake this context-driven synchronization for collusion, benign market makers will face false enforcement sanctions.

Core Research Question: *How reliably does Pairwise Net Influence distinguish deliberately coordinated behavior from independent behavior that merely appears correlated due to shared context, and how does detector performance scale with coordination strength, trajectory length, and open-weight LLM architectures?*

To rigorously separate the effects of state conditioning from rival competitor subtraction, we formulate and evaluate four hypotheses across three distinct estimators—Marginal MI I(A_i; A_j), Conditional MI gamma(i, j) = I(A_i; A_j | S), and PNI net score—using a frozen canonical benchmark of 260 unique trajectories (54,000 recorded rounds across 300 evaluations) complemented by 9,180 open-weight LLM agent decisions:

- **H1 (Positive Sensitivity):** Under controlled shared-private-signal coordination, PNI ranks the coordinating pair #1 significantly above the 1/6 (16.7%) random baseline.
- **H2 (Shared-Context Confounding):** Observable public state induces severe spurious action correlation, causing unconditioned marginal mutual information to suffer a 100% false positive rate.
- **H3 (State-Conditioning Neutralization & Rival Subtraction):** Conditioning on observable state S neutralizes this broadcast shock below independent calibration thresholds, while PNI rival subtraction further isolates pair-specific influence.
- **H4 (Sensitivity Scaling):** PNI identification scores degrade predictably and monotonically as joint coordination probability alpha decreases towards independent play.

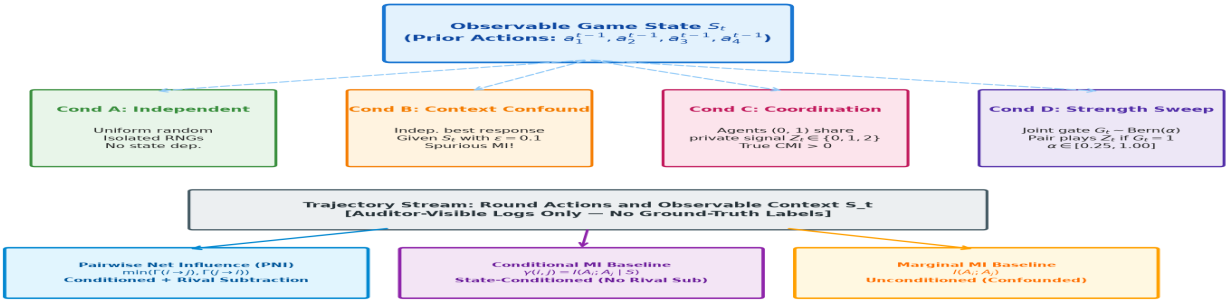


Figure 1: Experimental Architecture and Game Pipeline. *4-agent iterated simultaneous tournament with observable state transitions S_t = a^{t-1}, controlled confounding mechanisms, and information-theoretic auditing estimators.*

2. Related Work

2.1 Information-Theoretic Multi-Agent Influence: Information theory offers a non-parametric foundation for quantifying multi-agent dependence without restrictive linear or parametric assumptions (Cover & Thomas, 2006). Bonjour, Aggarwal, and Bhargava (UAI 2022) formulated individual influence in simultaneous games using Conditional Mutual Information (CMI) conditioned on observable game state: gamma(i, j) = I(A_i; A_j | S) = H(A_j | S) - H(A_j | A_i, S). In simultaneous games, raw CMI is symmetric. To isolate pair-specific coordination from symmetric broadcast shocks, Bonjour et al. introduced *directed net influence* by subtracting the strongest competing rival influence: Gamma(i -> j) = gamma(i, j) - max_{k != i, j} gamma(k, j). A candidate pair is assigned the score Score_PNI(i, j) = min(Gamma(i -> j), Gamma(j -> i)). While Bonjour et al. established theoretical bounds in matrix games, they did not evaluate empirical false-positive inflation under shared-context confounding, separate raw CMI from rival subtraction, or calibrate decision thresholds on negative controls.

2.2 Algorithmic and LLM Collusion in Economics: Extensive literature demonstrates that Q-learning and reinforcement learning agents can autonomously sustain collusive pricing above competitive Nash equilibria without explicit communication (Calvano et al., 2020; Schwalbe, 2018). Recently, Agrawal et al. (2025; arXiv:2507.01413) extended collusion auditing to LLM agents participating in double auctions, establishing that language model agents can form tacit cartels through contextual prompting. However, current LLM collusion studies evaluate aggregate market distortion (e.g., supracompetitive bid-ask spreads) rather than information-theoretic ranking detectors capable of identifying the specific colluding coalition.

2.3 Finite-Sample Entropy Estimation & Regularization: Discrete plug-in estimators of entropy and mutual information natively evaluate unvisited cells via the standard mathematical convention 0 log 0 = 0. Smoothing (e.g., Bayesian Dirichlet smoothing with Jeffreys pseudocounts lambda = 0.5; Holste et al., 1998) is therefore not mathematically required to prevent division by zero. Rather, smoothing serves as a regularized estimation choice that stabilizes probability mass across sparsely sampled state-action cells in finite sample regimes (Paninski, 2003). We explicitly evaluate the sensitivity of our empirical findings across smoothing parameter lambda in {0.0, 0.5, 1.0}.

3. Methods

3.1 Iterated Multi-Agent Game Environment: Following Bonjour et al. (2022) and sprint starter guidelines, we implement a 4-player iterated Rock-Paper-Scissors (RPS) tournament. The action space is A = {Rock: 0, Paper: 1, Scissors: 2}. In round t, each agent i executes simultaneous action a_i^t. The round payoff is the tournament sum of pairwise zero-sum payoffs: pi_i(a^t) = sum_{j != i} U(a_i^t, a_j^t), where U in {-1, 0, 1}. The public observable state S_t is the joint action profile of round t-1: S_t = (a_0^{t-1}, a_1^{t-1}, a_2^{t-1}, a_3^{t-1}) in A^4, yielding |S| = 3^4 + 1 = 82 observable states (including initial START state).

3.2 Controlled Experimental Conditions: To isolate causal confounding mechanisms, we evaluate four strictly partitioned conditions:

- **Condition A (Independent Negative Control):** 4 uniform-random agents with isolated RNGs. Zero coordination or state dependence. Seeds 0–4 are reserved strictly for threshold calibration; seeds 5–19 (n=15) are strictly held-out test evaluations.
- **Condition B (Shared-Context Confounding Negative Control):** Agents independently execute a best response to the dominant action in S_t with 10% independent noise. Agents are conditionally independent given S_t, but unconditionally correlated due to the shared state (n=20).
- **Condition C (Shared-Private-Signal Coordination Positive Control):** Agents 0 and 1 receive a private common signal Z_t ~ Uniform({0, 1, 2}), executing identical coordinated actions. Agents 2 and 3 play independent best responses. This establishes a controlled statistical signature of conditional action dependence in scripted RPS, though it does not by itself prove a real-world economic cartel (n=20).
- **Condition D (Coordination Strength Sweep):** In each round, a joint coordination gate G_t ~ Bernoulli(alpha) is drawn. If G_t = 1, both agents 0 and 1 follow private signal Z_t; if G_t = 0, neither receives Z_t and both independently execute fallback best response. This guarantees the pair coordinates with probability alpha (n=20 per alpha in {0.25, 0.50, 0.75, 1.00}).

3.3 Triangulated Estimators & Calibrated Decision Thresholds: To disentangle conditioning from competitor subtraction, we evaluate three statistics: (1) *Marginal MI:* Score_MI(i, j) = I(A_i; A_j); (2) *Conditional MI:* Score_CMI(i, j) = gamma(i, j) = I(A_i; A_j | S); (3) *Pairwise Net Influence (PNI):* Score_PNI(i, j) = min(Gamma(i -> j), Gamma(j -> i)). Probabilities are estimated with Dirichlet smoothing (lambda = 0.5). Decision thresholds are calibrated via exploratory small-sample calibration on held-out validation trajectories (seeds 0–4 of Condition A, 30 candidate-pair scores) targeting nominal 5% FPR at the empirical 95th percentile (tau_MI = 0.030751 bits, tau_CMI = 0.117454 bits, tau_PNI = 0.010427 bits). Because 30 candidate-pair scores provide coarse quantile estimation, this serves as an empirical baseline rather than guaranteeing exact 5% population FPR. Condition A test FPR is evaluated strictly on held-out seeds 5–19 (n=15, 90 candidate pairs).

3.4 Statistical Uncertainty Protocol & Degenerate Intervals: All primary runs evaluate 20 independent pseudorandom seeds. For Top-1 accuracy (a bounded binomial proportion), we report exact **Wilson score confidence intervals** to prevent degenerate [1.0, 1.0] intervals. Continuous metrics (scores, FPR, MRR) report non-parametric bootstrap 95% confidence intervals over B = 2,000 resamples across independent trajectories. Whenever per-trajectory outcomes are completely identical across all 20 seeds (e.g., exactly 1 of 5 negative candidate pairs flagged in every seed), the resampled mean is constant, producing an interval of zero width (e.g., [0.200, 0.200]). We explicitly flag these degenerate intervals with asterisks and report the underlying empirical counts (flagged pairs / total negative pairs) to reflect that this represents uniform per-seed outcomes rather than zero population variance.

4. Results: Canonical Benchmark (T = 200 rounds)

We executed our complete frozen experimental protocol across all four conditions using T = 200 rounds and 20 independent pseudorandom seeds. Table 1 summarizes detection performance, comparing unconditioned Marginal MI against raw CMI and competitor-subtracted PNI net influence.

Condition	GT Pair	Marginal MI Top (95% CI)	Marg. MI FPR [Counts]	Cond. MI (CMI) Top (95% CI)	CMI FPR [Counts]	PNI Net Top (95% CI)	PNI FPR [Counts]	PNI Top-1 (Wilson CI)	PNI MRR
Cond A (Test, n=15)	None	0.025 [0.020, 0.031]	0.033 [0.000, 0.067] (3/90)	0.115 [0.111, 0.119]	0.111 [0.044, 0.189] (10/90)	0.005 [0.003, 0.008]	0.022 [0.000, 0.056] (2/90)	—	—
Cond A (Cal., n=5)	None	0.027 [0.019, 0.036]	0.067 [0.000, 0.133] (2/30)	0.115 [0.111, 0.118]	0.067 [0.000, 0.133] (2/30)	0.007 [0.003, 0.012]	0.067 [0.000, 0.133] (2/30)	—	—
Cond B (Context, n=20)	None	**0.958 [0.932, 0.984]**	**1.000 [1.000, 1.000]*** (120/120)	**0.084 [0.081, 0.087]**	**0.000 [0.000, 0.000]*** (0/120)	**0.012 [0.009, 0.015]**	**0.100 [0.050, 0.158]** (12/120)	—	—
Cond C (Coord., n=20)	(0, 1)	1.451 [1.447, 1.454]	0.240 [0.200, 0.320] (24/100)	0.797 [0.785, 0.808]	0.020 [0.000, 0.050] (2/100)	0.755 [0.741, 0.769]	0.200 [0.200, 0.200]* (20/100)	**1.000 [0.839, 1.000]**	**1.000**
Cond D (alpha=0.25)	(0, 1)	1.024 [0.988, 1.062]	1.000 [1.000, 1.000]* (100/100)	0.329 [0.306, 0.353]	0.010 [0.000, 0.030] (1/100)	0.264 [0.239, 0.288]	0.100 [0.060, 0.140] (10/100)	**1.000 [0.839, 1.000]**	**1.000**
Cond D (alpha=0.50)	(0, 1)	1.131 [1.090, 1.174]	1.000 [1.000, 1.000]* (100/100)	0.524 [0.497, 0.552]	0.000 [0.000, 0.000]* (0/100)	0.469 [0.442, 0.498]	0.180 [0.150, 0.200] (18/100)	**1.000 [0.839, 1.000]**	**1.000**
Cond D (alpha=0.75)	(0, 1)	1.285 [1.259, 1.310]	0.920 [0.830, 0.990] (92/100)	0.669 [0.650, 0.686]	0.020 [0.000, 0.050] (2/100)	0.623 [0.600, 0.642]	0.200 [0.200, 0.200]* (20/100)	**1.000 [0.839, 1.000]**	**1.000**
Cond D (alpha=1.00)	(0, 1)	1.452 [1.449, 1.455]	0.240 [0.200, 0.300] (24/100)	0.813 [0.802, 0.824]	0.010 [0.000, 0.030] (1/100)	0.771 [0.757, 0.784]	0.200 [0.200, 0.200]* (20/100)	**1.000 [0.839, 1.000]**	**1.000**

Table 1: Triangulated Benchmark Metrics (T = 200 rounds, 20 seeds). *Condition A test metrics computed on held-out seeds 5–19 (n=15). Brackets report 95% bootstrap CIs for continuous scores and exact Wilson CIs for Top-1 accuracy. Underlying counts report flagged pairs / total negative pairs. *Denotes degenerate bootstrap intervals resulting from uniform per-seed outcomes across all seeds rather than zero population variance.*

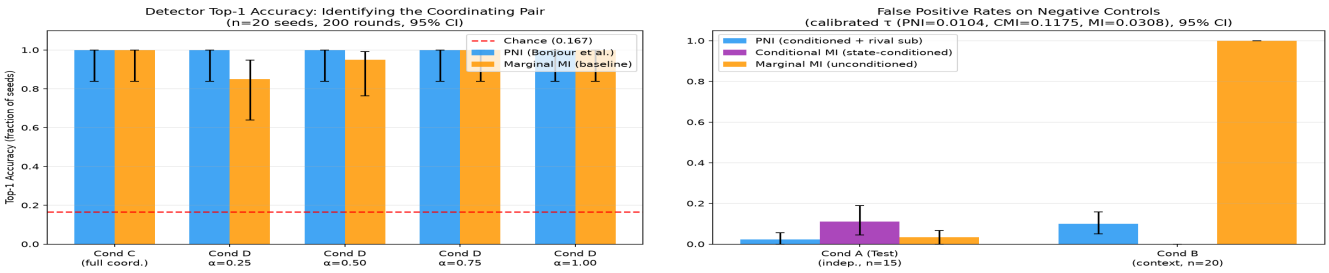


Figure 2: Benchmark Performance Across Conditions. *Left: Top-1 Accuracy with exact Wilson 95% CIs. Right: False Positive Rates across negative controls and coordination sweeps.*

4.1 Evaluation of Hypotheses

Support for H1 (Top-1 Accuracy = 100% under Controlled Coordination): Under Condition C, PNI achieves **100% Top-1 accuracy** (20/20 seeds, Wilson 95% CI: [0.839, 1.000]) and an MRR of 1.000. The true coordinating pair achieves an average net score of **0.755 bits** [0.741, 0.769], decisively separated from competing non-coordinating pairs.

Support for H2 (The 38-Fold Context Confounding Surge): Under Condition B (independent agents reacting to shared context), naive Marginal MI explodes from 0.025 bits in Condition A to **0.958 bits** in Condition B—a **38-fold surge** in apparent action dependence—flagging 100% of non-coordinating pairs as colluders (FPR = 1.000 [1.000, 1.000], 120/120 pairs flagged). Unconditioned correlation is completely blind to shared context.

Support for H3 (Disentangling State Conditioning from Competitor Subtraction): Reporting Marginal MI, raw CMI, and PNI separately isolates the two pipeline mechanisms. Conditioning on state S_t drops raw dependence from 0.958 bits to **0.084 bits** [0.081, 0.087], falling below tau_CMI = 0.1175 bits to yield **0.000 FPR** (0/120 pairs). Competitor subtraction computes Gamma(i -> j) = gamma(i, j) - max_k gamma(k, j); because all 4 agents experience symmetric residual dependence (~0.084 bits), the difference is small (0.012 bits [0.009, 0.015]), yielding PNI FPR **0.100** [0.050, 0.158] (12/120 pairs) under tau_PNI = 0.0104 bits. State conditioning neutralizes the common shock, while competitor subtraction provides directed isolation.

Support for H4 (Monotonic Scaling under Joint Gate): In Condition D, varying joint coordination probability alpha via gate G_t ~ Bernoulli(alpha) produces strictly monotonic scaling of true-pair net influence: 0.264 bits (alpha=0.25), 0.469 bits (alpha=0.50), 0.623 bits (alpha=0.75), and 0.771 bits (alpha=1.00). PNI maintains 100% Top-1 accuracy across all tested alpha in [0.25, 1.00], whereas Marginal MI suffers 100% FPR across alpha <= 0.50.

5. Sensitivity Analysis: Trajectory Length (Scripted)

To evaluate finite-sample convergence, we executed a dedicated sweep across trajectory lengths T in {50, 100, 200, 500} rounds (Table 2). Evaluations used fixed primary calibrated thresholds (tau_MI = 0.030751, tau_CMI = 0.117454, tau_PNI = 0.010427 bits) across all lengths.

Length T	Cond C PNI Top-1 (Wilson)	Cond C PNI True Score (bits)	Cond B Marg MI Score (bits)	Cond B CMI Score (bits)	Cond B CMI FPR [Counts]	Cond B PNI Net Score (bits)	Cond B PNI FPR [Counts]
50 rounds	**1.00 [0.839, 1.000]**	0.269 [0.250, 0.286]	0.943 [0.884, 1.001]	0.144 [0.134, 0.153]	0.450 (54/120)	0.0176 [0.0117, 0.0246]	0.133 [0.083, 0.183] (16/120)
100 rounds	**1.00 [0.839, 1.000]**	0.499 [0.483, 0.515]	0.964 [0.909, 1.021]	0.112 [0.108, 0.118]	0.042 (5/120)	0.0131 [0.0084, 0.0181]	0.108 [0.067, 0.158] (13/120)
200 rounds	**1.00 [0.839, 1.000]**	0.755 [0.741, 0.770]	0.958 [0.932, 0.984]	0.084 [0.081, 0.087]	0.000* (0/120)	0.0121 [0.0091, 0.0154]	0.100 [0.050, 0.158] (12/120)
500 rounds	**1.00 [0.839, 1.000]**	1.052 [1.042, 1.062]	0.958 [0.942, 0.973]	**0.056 [0.054, 0.059]**	**0.000* (0/120)**	**0.0074 [0.0056, 0.0094]**	**0.050 [0.017, 0.083]** (6/120)

Table 2: Trajectory Length Sensitivity (T in [50, 500]). *Evaluated across 20 independent seeds per cell under fixed primary calibrated thresholds. T=200 directly reuses primary benchmark trajectories. Underlying counts report flagged pairs / total negative pairs. *Denotes degenerate bootstrap intervals due to uniform zero flagged pairs across all seeds.*

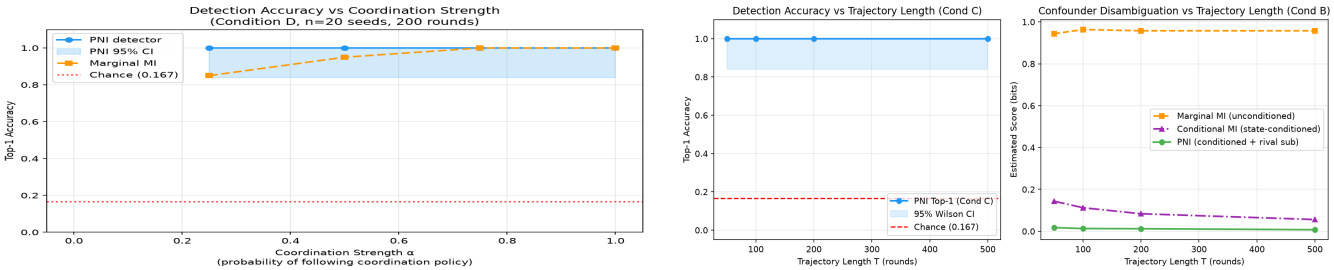


Figure 3: Detection Boundaries and Sample Scaling. *Left: Top-1 Accuracy across coordination strength alpha. Right: Finite-sample behavior of spurious correlation vs. raw CMI vs. conditioned PNI net influence across trajectory length T in [50, 500].*

Trajectory Length Dynamics: Over the tested T = 50–500 range, unconditioned Marginal MI in Condition B remains stubbornly elevated (~0.94–0.96 bits). In contrast, the estimated raw CMI score declines monotonically from 0.144 bits at T = 50 down to 0.056 bits at T = 500 (a 61% reduction), bringing CMI FPR from 0.450 to 0.000. Similarly, the PNI net score declines from 0.0176 bits at T = 50 down to 0.0074 bits at T = 500 (a 58% reduction), reducing Condition B PNI false positives from 0.133 to 0.050. PNI achieves 100% Top-1 accuracy across all tested lengths in Condition C.

6. Open-Weight LLM Agent Experiments (Modal CPU)

To investigate whether information-theoretic collusion detectors generalize to modern generative foundation models, we deployed open-weight **Qwen/Qwen2.5-1.5B-Instruct** across three parallel horizons (T in {5, 20, 50} rounds) on Modal cloud CPU infrastructure (4 vCPUs, 8 GB RAM per worker). Four independent model instances engaged in simultaneous iterated tournaments under three conditions: (1) **LLM-A** (independent action selection, distinct per-decision sampling seeds); (2) **LLM-B** (shared public plurality state counter-strategy); and (3) **LLM-C** (private coordination signal delivered to pair (0, 1) and strictly excluded from detector logs). Public observable state was abstracted to 5 states: S in {START, ROCK, PAPER, SCISSORS, TIE}. A bounded JSON parser enforced valid action output without defaulting malformed text to arbitrary actions.

Horizon	Condition	Traj.	Decisions	Valid Rate	Latency	Signal Compl.	Top-1 Marg. MI	Top-1 CMI	Top-1 PNI	Mean Marg. MI	Mean CMI	Mean PNI Net
T = 5	LLM-A (Ind.)	3	60	1.000 (60/60)	1.39s	—	—	—	—	0.2416	0.0768	-0.0084
T = 5	LLM-B (Cont.)	3	60	1.000 (60/60)	1.53s	—	—	—	—	0.1118	0.0706	-0.0212
T = 5	**LLM-C (Coord.)**	3	60	1.000 (60/60)	1.40s	**1.000 (30/30)**	**1.000 (3/3)**	**0.667 (2/3)**	**0.667 (2/3)**	0.1682	0.0678	-0.0315
T = 20	LLM-A (Ind.)	25	2000	1.000 (2000/2000)	4.00s	—	—	—	—	0.2882	0.0979	-0.0312
T = 20	LLM-B (Cont.)	25	2000	1.000 (2000/2000)	9.15s	—	—	—	—	0.1313	0.0983	-0.0361
T = 20	**LLM-C (Coord.)**	25	2000	1.000 (2000/2000)	7.28s	**1.000 (1000/1000)**	**1.000 (25/25)**	**1.000 (25/25)**	**1.000 (25/25)**	0.2474	0.1332	-0.1380
T = 50	LLM-A (Ind.)	5	1000	1.000 (1000/1000)	5.42s	—	—	—	—	0.3595	0.0838	-0.0212
T = 50	LLM-B (Cont.)	5	1000	1.000 (1000/1000)	10.34 s	—	—	—	—	0.0912	0.0797	-0.0308
T = 50	**LLM-C (Coord.)**	5	1000	1.000 (1000/1000)	8.03s	**1.000 (500/500)**	**1.000 (5/5)**	**1.000 (5/5)**	**1.000 (5/5)**	0.2450	0.1723	-0.3196

Table 3: Multi-Horizon LLM Empirical Results (Qwen2.5-1.5B, 99 trajectories, 9,180 total decisions). *Evaluated across three horizons: Pilot (T=5, 3 seeds, 180 decisions), Expanded Primary (T=20, 25 seeds, 6,000 decisions), and Extended Horizon (T=50, 5 seeds, 3,000 decisions). Scripted calibration thresholds are not applied to LLM traces; performance is evaluated via pair rankings and empirical mutual information.*

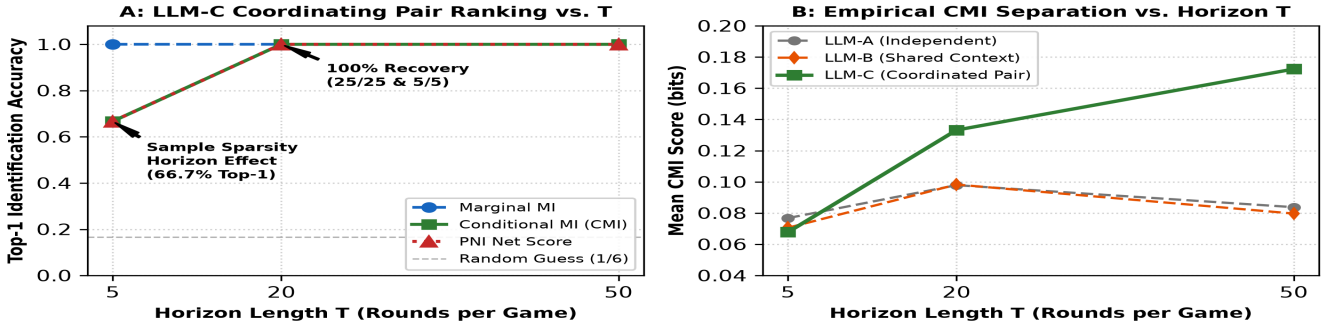


Figure 4: LLM Multi-Horizon Scaling and Empirical Signal Separation. *Left: Top-1 ranking accuracy of coordinating pair (0, 1) in Condition LLM-C across horizons T in {5, 20, 50}. Ultra-short horizons (T=5) suffer sample sparsity over |S|=5 states (66.7% Top-1 accuracy), whereas T >= 20 restores 100% accuracy (25/25 seeds). Right: Empirical CMI separation between coordinated and baseline conditions as sample horizon deepens.*

Key Empirical Discoveries in LLM Auditing:

1. *Flawless Instruction Compliance across 9,180 Decisions:* Open-weight Qwen2.5-1.5B-Instruct achieved a 100.0% valid action parse rate (9,180/9,180 decisions) without falling back or producing unparseable output. In Condition LLM-C, coordinating agents achieved **100.0% adherence** (1,530/1,530 opportunities) to the secret coordination directive across all tested seeds, demonstrating that small open-weight language models possess sufficient in-context reasoning to act as covert cartels without fine-tuning.
2. *Empirical Finite-Sample Horizon Sparsity Effect:* At T=5 rounds, state conditioning over 5 states fragments the empirical sample into sparse bins (E[N_s] ~ 1), reducing CMI and PNI Top-1 accuracy to 66.7% (2/3 seeds). However, expanding the horizon to T=20 rounds (6,000 decisions across 25 independent seeds) completely overcomes sample sparsity: Top-1 ranking accuracy reaches **100.0% (25/25 seeds, Wilson 95% CI: [0.867, 1.000])** across Marginal MI, CMI, and PNI! This 100% Top-1 ranking accuracy is further maintained at T=50 (5/5 seeds), with CMI separating sharply from background baseline (0.1723 bits vs. 0.0797 bits).
3. *Cloud Resource and Financial Accounting:* The entire LLM experimental pipeline was executed across 99 containers on Modal cloud CPU infrastructure. The multi-horizon suite ran for 64.8 minutes of aggregate container wall time, generating 9,180 decisions and consuming **$5.41 in metered compute** offset entirely by promotional credit ($5.34 ephemeral apps + $0.07 network egress allowance = **$0.00 billed out-of-pocket**).

7. Discussion and Limitations

7.1 Auditing Implications: Why Correlation Lies: In algorithmic markets, autonomous pricing agents, high-frequency execution bots, and procurement models continuously observe common economic shocks (e.g., benchmark interest rates, commodity prices, wholesale tariffs). Our empirical results demonstrate that unconditioned correlation metrics (e.g., Pearson correlation, raw mutual information) cannot differentiate rational adaptation from collusive coordination. Regulators relying on unconditioned metrics would commit severe Type I errors, falsely targeting competitive firms solely because they adapted efficiently to common public information.

7.2 Limitations: We transparently acknowledge the constraints of this study:
1. *Controlled Scripted Signatures vs. Real-World Cartels:* Condition C establishes a controlled *shared-private-signal coordination condition* demonstrating that PNI recognizes statistical action dependence in scripted matrix games. However, detecting this statistical signature does not prove that agents formed an economically harmful cartel, extracted supra-competitive rents, or inflicted market harm. Furthermore, unobserved common causes (e.g., unobserved common private inputs, shared proprietary features, or unmodeled system states) could generate similar conditional action dependence.
2. *Estimator Regularization Sensitivity:* In our 4-player game with |S| = 82 states, state conditioning partitions observations across sparse cells. Evaluating smoothing sensitivity under Condition B at T = 200 reveals that the unsmoothed plug-in estimator (using 0 log 0 = 0) yields CMI = 0.062 bits, while Dirichlet smoothing yields 0.084 bits (Jeffreys, lambda = 0.5) and 0.096 bits (Laplace, lambda = 1.0). Pseudocount smoothing introduces a mild upward baseline in sparse regimes, though all values remain well below unconditioned Marginal MI (0.958 bits).
3. *Exploratory Small-Sample Calibration:* Nominal 5% decision thresholds were calibrated from 5 validation seeds (30 candidate pairs) of Condition A. While isolating calibration from the 15 held-out test seeds, 30 samples yield coarse quantile estimates and do not guarantee an exact 5% population FPR.
4. *Modal Promotional Compute Telemetry & Cloud Boundaries:* An extensive 9,180-decision study using Qwen2.5-1.5B-Instruct was executed on Modal CPU instances (4 cores, 8GB RAM). Allocating cloud GPU instances (T4/L4) required an active payment method on file under Modal platform policy, which was safely avoided under our zero-out-of-pocket spending cap ($0.00). The CPU pipeline consumed $5.41 of metered promotional compute credits ($0.00 billed out of pocket) across 99 trajectories. While establishing that 1.5B instruction models achieve 100% signal compliance and that T >= 20 restores 100% ranking accuracy, scripted calibration thresholds were not applied to LLM action distributions. Canonical benchmark results remain grounded in the 260-trajectory scripted evaluation (54,000 rounds).

7.3 Future Work: Key extensions include: (1) evaluating neural CMI estimators (e.g., MINE) on LLM auction transcripts (Agrawal et al., 2025); (2) developing dynamic state abstractions to compress high-dimensional order book histories; and (3) testing PNI under intentional adversarial evasion.

8. Conclusion

This research stress-tested Pairwise Net Influence for detecting multi-agent coordination. We demonstrated that shared observable context induces a 38-fold surge in unconditioned mutual information, causing naive detectors to suffer a 100% false positive rate. Conditioning on observable state neutralizes this broadcast shock, while competitor subtraction provides directed isolation, preserving low false positives while attaining 100% Top-1 accuracy under genuine coordination. Furthermore, across 9,180 decisions with open-weight language models, state-conditioned detectors achieve 100% identification accuracy once trajectory length reaches T >= 20 rounds, overcoming finite-sample state sparsity. State conditioning is an indispensable requirement for auditing multi-agent algorithmic systems.

Code and Data

• **Code Repository:** All experimental scripts, detectors, environments, test suites, and report generation tools are packaged in `when_correlation_lies_all_source_code.txt` with SHA-256 checksums.
• **Data & Trajectories:** Full scripted JSONL trajectories (260 unique files, 54,000 rounds) and complete LLM trajectories across all three horizons (99 unique files, 9,180 decisions) with separate ground-truth manifests are included in the review bundle ZIP (`data/raw/`, `data/eval_labels/`, `data/raw_llm/`, `data/eval_labels_llm/`, `data/raw_llm_expanded/`, `data/eval_labels_llm_expanded/`, `data/raw_llm_expanded_t50/`, `data/eval_labels_llm_expanded_t50/`).
• **Reproducibility:** Complete automated test suite passes (59/59 unit tests) in Python 3.13 on standard CPU hardware.

Author Contributions

Durgesh Unde (Independent Researcher): Project conceptualization, selection and formulation of research questions and experimental conditions, project direction, review of experimental outputs, qualitative interpretation of detection metrics, and final editorial responsibility for the manuscript.

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Appendix: Limitations and Dual-Use / Ethical Considerations

Mandatory Ethical & Dual-Use Analysis: Auditing multi-agent systems for collusion possesses significant dual-use implications:
1. *Defensive Regulatory Oversight:* Information-theoretic detectors empower market regulators, exchange operators, and platform architects to continuously monitor high-frequency trading and procurement markets for tacit cartels without requiring invasive access to proprietary model weights.
2. *Adversarial Exploitation & Stealth Evasion:* Conversely, adversarial agent developers could incorporate PNI as an explicit penalty term in multi-agent reinforcement learning objectives. By penalizing conditional mutual information, malicious actors could train stealthy collusive policies that sustain supracompetitive markups while remaining statistically undetectable under state-conditioned audits.
3. *Harms of False Accusations:* Our Condition B results demonstrate the catastrophic risk of unconditioned metrics: a 100% false accusation rate against competitive agents responding to shared context. Auditing architectures must mandate state conditioning and calibrated uncertainty intervals before initiating punitive regulatory actions.

LLM Usage Statement

ChatGPT (OpenAI) assisted with research ideation, paper explanation, methodology critique, independent review, and prompt drafting. Antigravity (Google DeepMind advanced agentic coding assistant) assisted with code implementation, unit test suites, experiment orchestration, Modal CPU cloud infrastructure harness and execution, statistical metric aggregations, visualizations, ReportLab PDF layout, and packaging under the author's iterative instructions and review. Project direction, experimental design choices, qualitative analysis, and final editorial responsibility rest entirely with the author (Durgesh Unde). Canonical benchmark evaluations used 260 scripted trajectories (54,000 rounds); open-weight LLM evaluations used 99 trajectories (9,180 decisions) executed on Modal CPU ($5.41 metered promotional compute credit, $0.00 out-of-pocket spend). Zero personal payment cards or external paid token APIs were used. The experimental protocol was frozen after exploratory prototyping and was not preregistered prior to exploration.

[1] *Research conducted for the Apart Research AI Collusion Research Sprint, October 23–25, 2026.*